

Rethinking models

Speedup and numerical differences
Prepared for Richard McElreath | 16 September 2026

We compared the unchanged Stan programs and processed data from all 61 `ulam()` call sites in the second-edition book supplement. Every call site completed four sampling seeds in both engines. A separate hurdle example is also included.

The median end-to-end sampling speedup was 1.58x (`CmdStan` time / `Stanli` time). All 61 book call sites had a ratio above 1. `Stanli` preparation is included; the `CmdStan` model was already compiled. A value of 2x means `Stanli` took half as long.

Numerical differences

At three fixed parameter vectors per fixture, we compared 32,349 values across 62 fixtures. The largest absolute differences were 1.88e-09 for log density, 1.86e-09 for a gradient component, and 3.55e-15 for a model output. The tables give the model-specific maxima.

Examples

Speedup is `CmdStan`/`Stanli` complete CLI time. Numerical columns show maximum absolute differences.

Book model / example	Sampling speedup	Log density difference	Gradient difference
m11.4 - Chimpanzee logistic regression	1.81x	0	3.20e-14
m12.5 - Ordered response with predictors	1.58x	5.86e-10	1.00e-11
m12.6 - Monotonic education effect	1.45x	8.37e-10	3.50e-12
m13.4nc - Non-centered multilevel model	2.62x	0	2.93e-14
m14.8 - Gaussian-process model	1.51x	5.82e-11	9.31e-10
m14.11 - Phylogenetic GP, 151 species	1.17x	0	3.21e-12
m15.2 - Measurement-error model	1.79x	1.14e-13	2.22e-16
m16.1 - Nonlinear height-weight model	2.35x	1.14e-13	1.71e-13

Estimated first-fit speedup

Including the separately measured `CmdStan` translation and C++ build gives a median estimated first-fit speedup of 37.32x across the 61 book call sites. These estimates add measured stages; they are not directly timed combined runs. Installation is excluded. The main tables use the already-compiled `CmdStan` comparison.

The appendix lists every book call site and the separate hurdle fixture. Sampler measurements and original times remain in the supporting data.

How to read the results

Sampling speedup

Sampling speedup = median CmdStan CLI time / median Stanli CLI time, using four seeds per engine. Each run has 1,000 warmup iterations and 1,000 retained draws. Stanli time includes Stan translation, preparation, warmup, sampling, generated quantities and CSV output. CmdStan time starts from its compiled model. The ratio describes this fixed sampling budget; it does not compare time to an equal effective sample size.

Absolute numerical differences

Absolute difference = |Stanli value - CmdStan value|. Log density includes the parameter-transform Jacobian. The gradient column takes the maximum over all unconstrained parameter derivatives; the output column covers constrained parameters, transformed parameters and generated quantities. Each maximum spans all three fixed parameter vectors. These are comparisons at identical parameter values, not differences between random posterior draws.

The supporting CSV also reports ULP distance: the number of representable double-precision steps between the two values. ULP distance depends on the scale of the values and can be large near zero despite a tiny absolute difference. The report therefore uses absolute differences; zero means the compared values were equal.

Measurement conditions

Apple M3 Ultra, 96 GiB RAM, macOS ARM64; Release build; one engine at a time. Rethinking 2.42, CmdStan 2.39.0 and posterior 1.7.0. Runtime/compiler revision 6e462c2e; run fd5e0ecacddc7047. The toolchain was already installed. The manifest records executable, dependency and input identities.

Seeds 1-4; target acceptance 0.8; maximum tree depth 10; random initialization. Every model used the same iteration budget. The book's model-specific tuning and chain-count choices were not reproduced. CmdStan ran first for each seed. Stanli had a limit of three times that CmdStan CLI duration or 900 seconds, whichever was smaller; CmdStan also had a 900-second limit. All book runs completed. Sampling order was not counterbalanced.

Supporting measurements

Numerical comparisons use the separately recorded high-precision CmdStan values and the frozen Stanli checker from the timing build. The retained per-seed CSVs use the CLI defaults: eight significant digits for CmdStan and 17 for Stanli. The supporting CSV lists divergence counts, maximum R-hat and minimum bulk ESS for each engine.

Scope and sources

All 61 `ulam()` call sites in the chapters 4-16 supplement are listed, representing 58 distinct Stan/data pairs. Repeated fits remain listed. The separate hurdle example brings the total to 62 fixtures. Chapters 4-8 and 10 have no `ulam()` calls in this supplement. `quap()`, direct `stan()` calls, exercises and lecture-only models are outside this inventory.

Pinned second-edition code supplement (rethinking 2.42)

The accompanying CSV and JSON retain every speedup, numerical comparison, sampler measurement and build/input identity. `numerical-values.json.gz` contains the paired values from both engines.

Full results

Speedup: CmdStan/Stanli median complete CLI time. Error columns: maximum absolute difference across three parameter vectors. The full-precision values, per-seed times, ULP distances and sampler measurements are retained in the accompanying CSV and JSON files.

Book model	R code	Sampling speedup	Log density max abs. diff.	Gradient max abs. diff.	Output max abs. diff.
m9.1	9.14	1.82x	0	2.84e-14	0
m9.1	9.16	1.84x	0	2.84e-14	0
m9.2	9.22	1.41x	0	0	0
m9.3	9.24	1.35x	0	0	0
m9.4	9.26	1.45x	0	0	0
m9.5	9.27	1.41x	0	0	0
mp	9.28	3.12x	0	0	0
m5.8s	9.29	2.89x	0	2.73e-12	0
m5.8s2	9.30	3.01x	0	1.36e-12	0
m11.4	11.11	1.81x	0	3.20e-14	0
m11.5	11.19	2.11x	0	1.03e-13	0
m11.6	11.25	1.42x	0	0	0
m11.7	11.29	1.27x	0	1.14e-13	0
m11.8	11.32	1.51x	0	0	0
m11.9	11.45	1.26x	0	0	0
m11.10	11.45	1.36x	0	0	0
m11.11	11.49	1.62x	0	1.86e-09	0
m_pois	11.61	1.14x	0	0	0
m12.1	12.2	1.28x	0	8.88e-16	0
m12.2	12.6	1.41x	1.42e-14	7.11e-15	0
m12.3	12.9	9.28x	1.99e-12	5.54e-13	0
m12.3_alt	12.11	4.42x	1.19e-12	9.95e-13	0
m12.4	12.16	146.87x	1.88e-09	1.42e-10	0
m12.5	12.24	1.58x	5.86e-10	1.00e-11	0
m12.6	12.34	1.45x	8.37e-10	3.50e-12	0
m12.7	12.37	1.60x	8.44e-10	7.19e-11	0
m13.1	13.2	1.49x	0	0	0
m13.2	13.3	1.49x	0	0	0
m13.3	13.13	1.12x	0	0	0
m13.4	13.21	1.61x	0	2.84e-14	0
m13.5	13.23	1.79x	0	4.97e-14	0

Full results (continued)

Speedup: CmdStan/Stanli median complete CLI time. Error columns: maximum absolute difference across three parameter vectors. The full-precision values, per-seed times, ULP distances and sampler measurements are retained in the accompanying CSV and JSON files.

Book model	R code	Sampling speedup	Log density max abs. diff.	Gradient max abs. diff.	Output max abs. diff.
m13.6	13.25	2.19x	0	2.84e-14	0
m13.7	13.26	1.32x	0	0	0
m13.7nc	13.27	1.28x	0	0	0
m13.4b	13.28	1.47x	0	2.84e-14	0
m13.4nc	13.29	2.62x	0	2.93e-14	5.55e-17
m14.1	14.12	1.40x	0	1.07e-14	0
m14.2	14.18	1.46x	5.68e-14	1.07e-14	0
m14.3	14.19	1.76x	5.68e-14	1.07e-14	0
m14.4	14.24	2.60x	0	1.42e-13	0
m14.5	14.25	3.31x	0	8.53e-14	0
m14.6	14.26	1.71x	1.14e-13	2.84e-13	0
m14.4x	14.27	2.61x	0	1.42e-13	0
m14.6x	14.27	1.74x	1.14e-13	2.84e-13	0
m14.7	14.31	1.45x	2.27e-13	5.91e-12	0
m14.8	14.39	1.51x	5.82e-11	9.31e-10	0
m14.8nc	14.46	1.44x	0	1.86e-09	0
m14.9	14.49	1.22x	0	5.68e-14	0
m14.10	14.51	1.24x	1.42e-14	6.39e-14	0
m14.11	14.52	1.17x	0	3.21e-12	0
m15.1	15.3	1.94x	0	1.78e-15	0
m15.2	15.5	1.79x	1.14e-13	2.22e-16	0
m15.3	15.12	1.15x	0	0	0
m15.4	15.13	1.32x	0	0	0
m15.5	15.17	1.76x	0	3.55e-15	0
m15.6	15.19	1.44x	0	1.78e-15	0
m15.7	15.22	1.65x	0	4.00e-15	0
m15.8	15.30	2.00x	1.99e-13	5.68e-14	0
m15.9	15.31	2.15x	1.99e-13	1.71e-13	3.55e-15
m16.1	16.2	2.35x	1.14e-13	1.71e-13	0
m16.4	16.11	2.83x	0	1.71e-13	0

Supplemental fixture (not a book model)

Hurdle Poisson: 1.29x speedup. Maximum absolute differences: 3.55e-15 (log density), 8.88e-16 (gradient), 0 (output).